Math 10 - Systems of Linear Equations



— SINE TIMORE AUT FAVORE —

ST. GEORGE'S SCHOOL

Name : _____

Teacher's name: _____

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Contents

1 Solving Systems by Graphing **4**
Solving with Technology 6
Practice Exercises 7

2 Solving Systems by Substitution **10**
Practice Exercises 12

3 Solving Systems by Elimination **16**
Practice Exercises 18

4 Applications of Systems of Equations **22**
Practice Exercises 25

5 Reasoning and Analyzing **28**
Practice Exercises 30

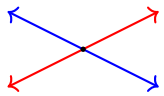
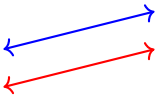
6 Answer Key **32**
Section 1 - Exercises 32
Section 2 - Exercises 32
Section 3 - Exercises 32
Section 4 - Exercises 32
Section 5 - Exercises 33

1 Solving Systems by Graphing

Solving a **system of equations** is trying to find one solution that satisfies all the given equations.

There are a few different ways that systems of equations are commonly solved.

When graphing two lines, one of three scenarios can occur.

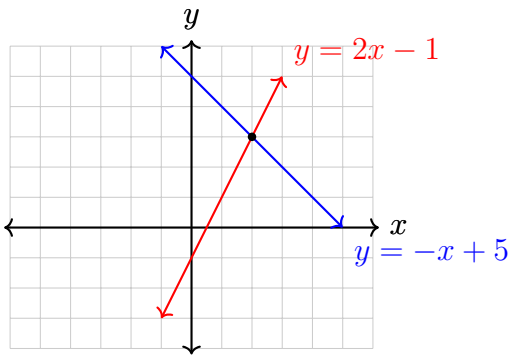
intersecting lines	parallel lines	coincident lines
		
one solution	no solutions	infinite solutions

When we graph by hand, we are limited to how accurate our answers can be. For the most part, we will have integer coordinates as solutions if graphing by hand. There is also technology that can help us graph as well, such as graphing calculators, Desmos, and Geogebra.

Skills Needed

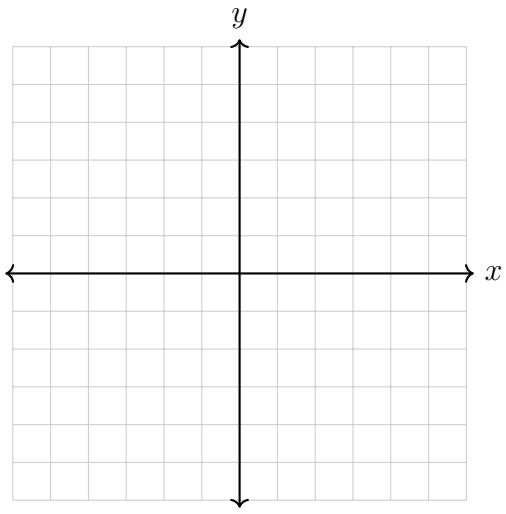
1. Sketching lines in $y = mx + b$ form
2. Sketching lines in $Ax + By = C$ form
3. Sketching lines in $y - y_1 = m(x - x_1)$ form
4. Sketching horizontal $(y = k)$ and vertical $(x = k)$ lines
5. Manipulating equations algebraically as needed

Example 1.1. Determine the solution to the system of linear equations shown in the graph below.

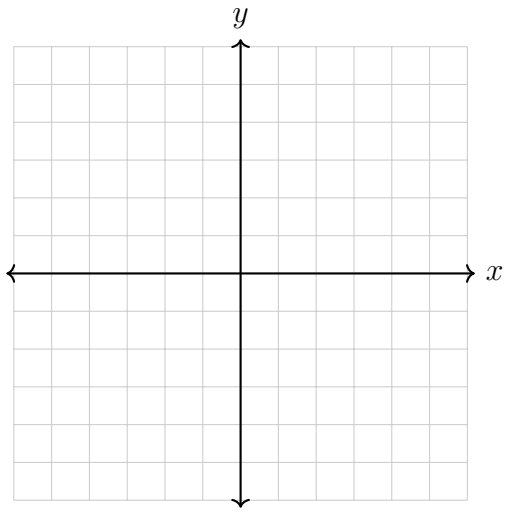


Example 1.2. Solve the following systems of equations graphically.

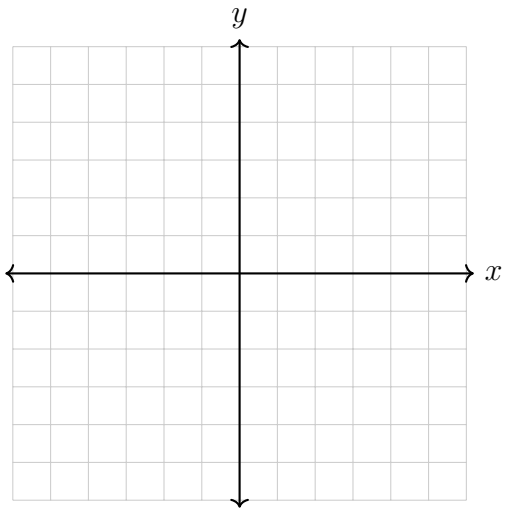
a) $\begin{cases} y = 2x - 4 \\ 2x + 3y = 12 \end{cases}$



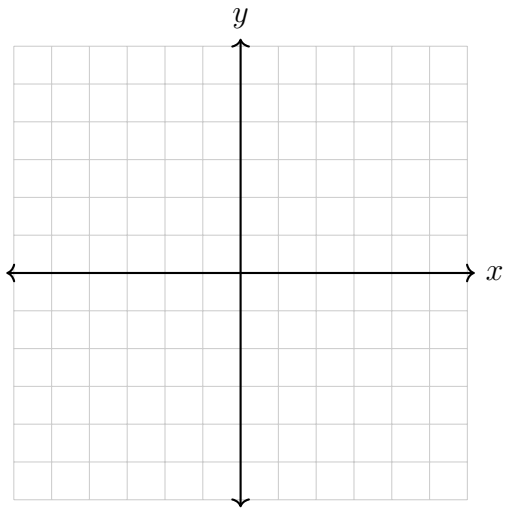
b) $\begin{cases} x + 2y = -4 \\ x - y = 5 \end{cases}$



c) $\begin{cases} y - 3 = (x + 2) \\ y + 3 = -2(x + 2) \end{cases}$



d) $\begin{cases} y = \frac{1}{2}x + 2 \\ x - 2y = 2 \end{cases}$

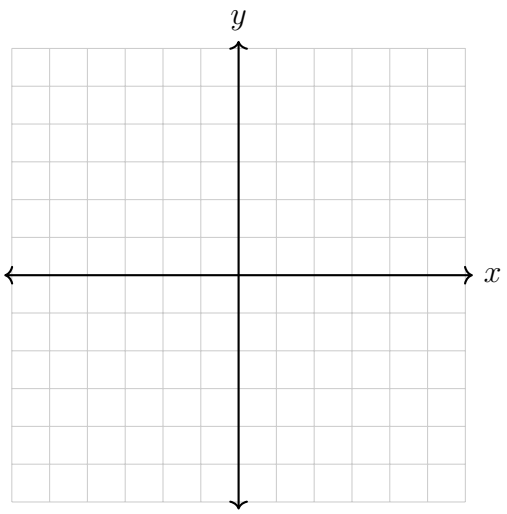


Solving with Technology

While graphing by hand is an excellent way to understand the concept of a solution, it has practical limitations. If the intersection point occurs at non-integer coordinates (fractions or decimals), or if the equations involve very large numbers, reading the exact solution from a hand-drawn graph becomes practically impossible.
In these cases, we must either *estimate* the solution visually or use graphing technology (like Desmos or a Graphing Display Calculator) to find the precise intersection.

Example 1.3. Graph the following system by hand. Estimate the solution based on your graph. Then, use graphing technology to find the exact coordinate.

$$\begin{cases} y = 3x - 2 \\ y = -4x + 1 \end{cases}$$



Example 1.4. Use graphing technology to solve the following systems involving larger numbers and decimals. Round to the nearest hundredth if necessary.

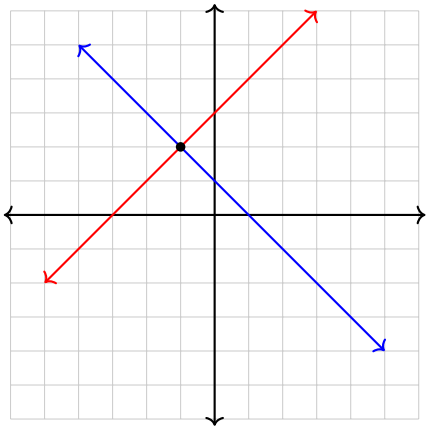
a) $\begin{cases} y = 15x - 250 \\ y = -8x + 320 \end{cases}$

b) $\begin{cases} 2.5x - 3.2y = 14.1 \\ -1.1x + 4.5y = -8.3 \end{cases}$

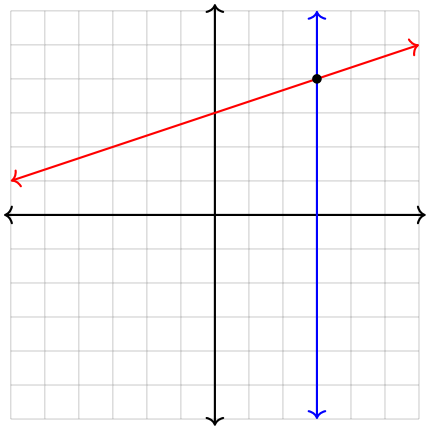
Practice Exercises

1. Read the solution to the system of equations from the given graphs.

a)

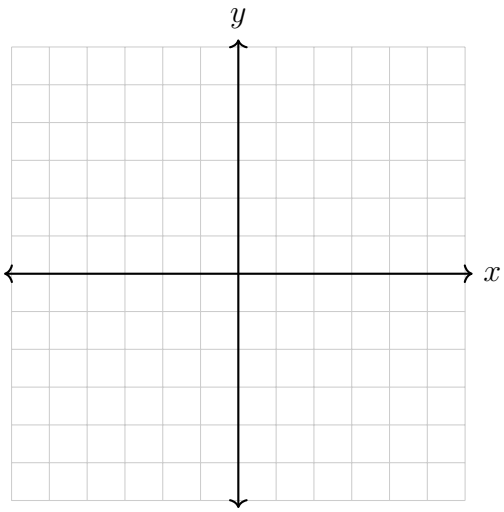


b)

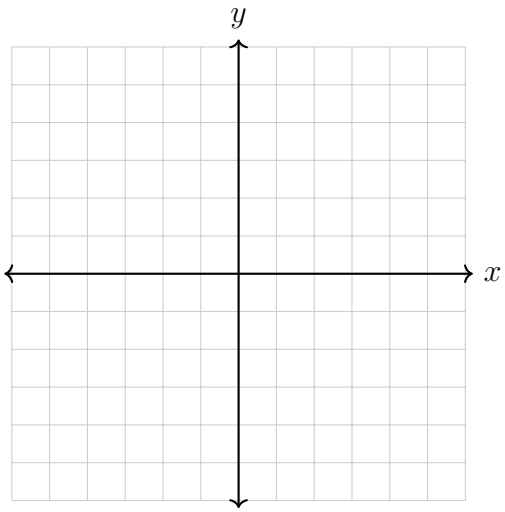


2. Solve for each system of equation by graphing, without using technology.

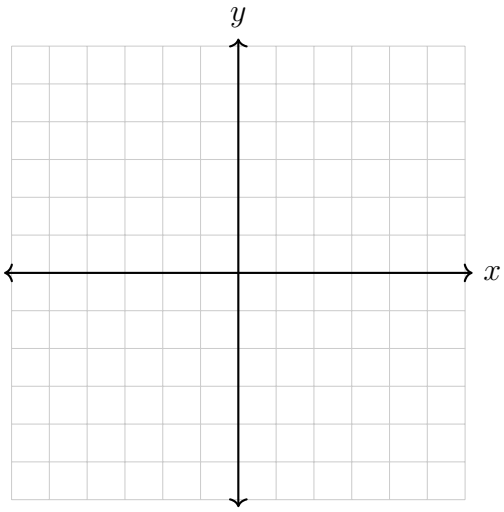
a) $\begin{cases} y = -3x + 4 \\ y = 3x - 2 \end{cases}$



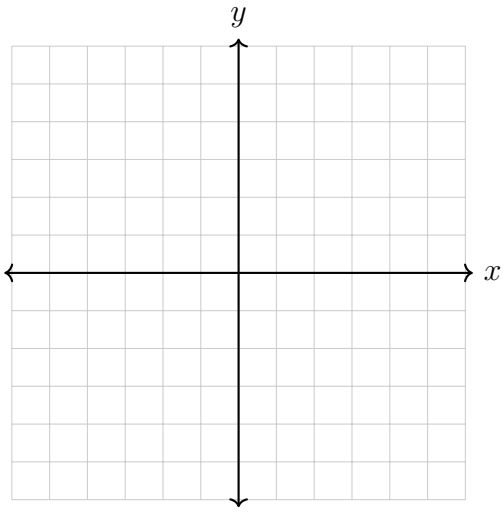
b) $\begin{cases} y = x + 2 \\ x = -3 \end{cases}$



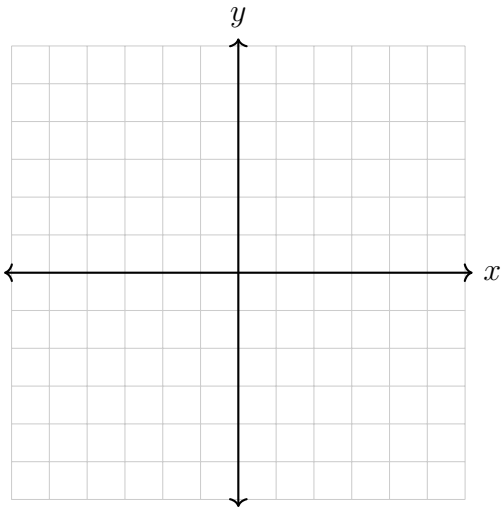
c) $\begin{cases} x - y = 3 \\ 7x - y = -3 \end{cases}$



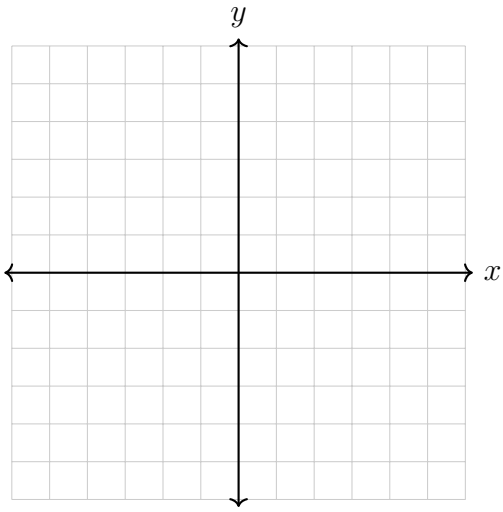
d) $\begin{cases} 4x + y = 2 \\ x - y = 3 \end{cases}$



e) $\begin{cases} y = -x \\ x + 2y = 6 \end{cases}$

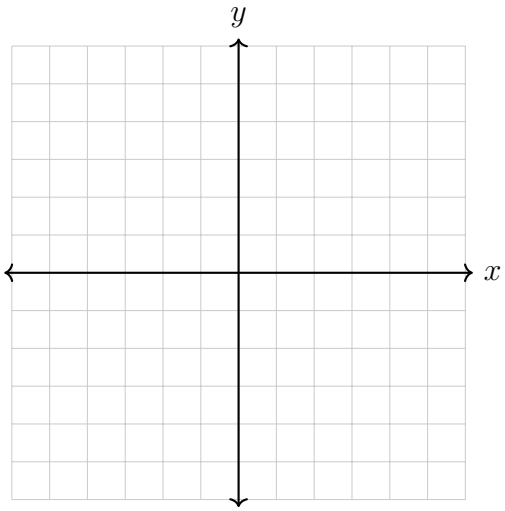


f) $\begin{cases} x = 3 - 3y \\ 2y = x + 2 \end{cases}$



3. Graph the following system by hand to estimate the point of intersection. Then, state the exact solution by using graphing technology.

$$\begin{cases} y = \frac{1}{2}x + 2 \\ y = -3x - 1 \end{cases}$$



4. Solve each system of equations by using graphing technology. Round your answers to the nearest hundredth where necessary.

a) $\begin{cases} y = 3x - 7 \\ y = -x + 9 \end{cases}$

b) $\begin{cases} y = -x \\ y = -\frac{1}{3}x + 3 \end{cases}$

c) $\begin{cases} 3x + 2y = 5 \\ x - y = 1 \end{cases}$

d) $\begin{cases} 4x - y = 6 \\ 4x + y = 1 \end{cases}$

e) $\begin{cases} y = 125x + 800 \\ y = -75x + 2400 \end{cases}$

f) $\begin{cases} 1.2x + 4.5y = 12 \\ -3.8x + y = -5.4 \end{cases}$

2 Solving Systems by Substitution

When graphing is not effective, or when technology is not available, algebraic methods are always available.

One method that works even beyond linear equations is **substitution.** This technique allows us to write an equation with just one variable.

Example 2.1. Solve by substitution.

a) $\begin{cases} y = x + 5 \\ 3x - y = 1 \end{cases}$

b) $\begin{cases} 4x - 3y = 2 \\ y = 2x + 1 \end{cases}$

c) $\begin{cases} x = 3y + 2 \\ x - 2y = 1 \end{cases}$

d) $\begin{cases} 2x + y = -3 \\ 3x + 2y = -4 \end{cases}$

Isolating Variables and Rational Coefficients

In the previous examples, one variable was either already isolated or had a coefficient of 1 or -1, making it easy to isolate without creating fractions. However, sometimes you will be forced to work with rational coefficients (fractions).
When choosing which variable to isolate, look for the one that will create the simplest fractions. Alternatively, if the original equations contain fractions or decimals, you may want to clear them first before substituting.

Example 2.2. Solve by substitution. Note that isolating any variable will result in fractional expressions.

a) $\begin{cases} 2x + 5y = 19 \\ 3x - 4y = -6 \end{cases}$

b) $\begin{cases} 3x + 2y = 14 \\ 5x - 3y = 17 \end{cases}$

Example 2.3. Solve by substitution. Clear the fractions or decimals first if it makes the process easier.

a) $\begin{cases} \frac{1}{2}x + \frac{2}{3}y = 1 \\ x - \frac{1}{3}y = 7 \end{cases}$

b) $\begin{cases} 0.4x - 1.2y = 2.8 \\ 1.5x + 0.5y = -2.5 \end{cases}$

Practice Exercises

1. Solve each of the following systems of equations by substitution.

a) $\begin{cases} y = x + 2 \\ 3x + 4y = 1 \end{cases}$

b) $\begin{cases} x - 2y = 10 \\ x + 5y = -4 \end{cases}$

c) $\begin{cases} 4p + q = 0 \\ 7p + 4q = 3 \end{cases}$

d) $\begin{cases} 5x - 3y = -11 \\ x - 2y = 2 \end{cases}$

e) $\begin{cases} x - y = 3 \\ 6x + 4y = 13 \end{cases}$

f) $\begin{cases} 2x - y = 16 \\ -x + 2y = 8 \end{cases}$

g) $\begin{cases} y = 3x - 7 \\ 6x - 2y = 14 \end{cases}$

h) $\begin{cases} x = 3y + 2 \\ 2x - 6y = 5 \end{cases}$

2. Solve each of the following systems by substitution. You will need to isolate a variable first, which may introduce fractions.

a) $\begin{cases} 2x + 3y = 12 \\ 5x - 2y = 11 \end{cases}$

b) $\begin{cases} 4x - 5y = 22 \\ 3x + 2y = -18 \end{cases}$

c) $\begin{cases} 3a + 5b = 16 \\ 4a - 3b = 12 \end{cases}$

d) $\begin{cases} 6m - 5n = 1 \\ 4m + 3n = 7 \end{cases}$

3. Solve the following systems of equations containing rational coefficients.

a) $\begin{cases} \frac{x}{3} + \frac{y}{2} = 4 \\ \frac{2x}{3} - \frac{y}{4} = 1 \end{cases}$

b) $\begin{cases} \frac{1}{4}x - \frac{1}{3}y = -1 \\ \frac{1}{2}x + \frac{1}{5}y = 5 \end{cases}$

c) $\begin{cases} 0.2x + 0.3y = 2.1 \\ 0.5x - 0.1y = 2.7 \end{cases}$

d) $\begin{cases} 1.5x - 0.2y = 3.4 \\ 0.3x + 0.8y = -1.4 \end{cases}$

3 Solving Systems by Elimination

Another technique that works well is the elimination method. This does not require the isolation of a variable and allows us to work with integer coefficients for the most part.

Elimination involves the process of manipulating the equations so that one variable has the exact same coefficient (or the exact opposite) in both equations. Then, we combine the equations to eliminate that variable entirely.

- **Opposite signs: Add** the equations together.
- **Same signs: Subtract** the equations. (Alternatively, multiply one equation by -1 and then add, which often helps prevent integer subtraction mistakes).

Example 3.1. Solve by elimination.

a) $\begin{cases} 4x + 3y = 2 \\ x - 3y = 8 \end{cases}$

b) $\begin{cases} 3x - 2y = -5 \\ 3x + y = -11 \end{cases}$

If the coefficients of the variables do not match in size, we can multiply one or both equations by a constant to create a match.

Example 3.2. Solve by elimination by multiplying one or both equations.

a) $\begin{cases} 3x + 2y = 7 \\ 2x - 5y = 11 \end{cases}$

b) $\begin{cases} 5x + 4y = 14 \\ 3x - 3y = 3 \end{cases}$

Rational Coefficients

Just like with substitution, if your system contains fractions or decimals, it is usually best to clear them first by multiplying the entire equation by the Lowest Common Denominator (LCD) or a power of 10. Once the coefficients are integers, proceed with standard elimination.

Example 3.3. Solve by elimination. Clear the fractions or decimals first.

a) $\begin{cases} \frac{1}{2}x - \frac{2}{3}y = -1 \\ \frac{1}{4}x + \frac{1}{2}y = 4 \end{cases}$

b) $\begin{cases} 0.3x + 0.4y = 2.5 \\ 0.2x - 0.3y = 0.5 \end{cases}$

Practice Exercises

1. Solve each of the following systems of equations by elimination. Verify that the solution satisfies both equations.

a) $\begin{cases} 8x - y = 10 \\ 4x + y = 14 \end{cases}$

b) $\begin{cases} 7x + y = 15 \\ 3x + y = 3 \end{cases}$

c) $\begin{cases} 2a + 5b = 16 \\ a - b = 1 \end{cases}$

d) $\begin{cases} 4x + 3y = 9 \\ 2x - 5y = 1 \end{cases}$

e) $\begin{cases} 2x + 4y = 7 \\ 4x - 3y = 3 \end{cases}$

f) $\begin{cases} 3x + 2y - 6 = 0 \\ 9x = 5y + 18 \end{cases}$

g) $\begin{cases} x - 2y + 1 = 0 \\ 2x + 3y = 12 \end{cases}$

h) $\begin{cases} 7x + 3y = 5 \\ 5x - 6y = 9 \end{cases}$

2. Solve each of the following systems by elimination. You will need to multiply both equations to create matching coefficients.

a) $\begin{cases} 3x - 5y = 14 \\ 2x + 3y = 3 \end{cases}$

b) $\begin{cases} 4x + 7y = -5 \\ 3x - 2y = 18 \end{cases}$

c) $\begin{cases} 5m - 3n = 21 \\ 2m + 4n = 24 \end{cases}$

d) $\begin{cases} 6p + 5q = 8 \\ 5p - 4q = -26 \end{cases}$

3. Solve the following systems of equations containing rational coefficients by using elimination.

a) $\begin{cases} \frac{2}{3}x - \frac{1}{2}y = 1 \\ \frac{1}{4}x + \frac{1}{3}y = 5 \end{cases}$

b) $\begin{cases} \frac{3}{5}x + \frac{2}{3}y = 2 \\ \frac{1}{2}x - \frac{3}{4}y = -3 \end{cases}$

c) $\begin{cases} 0.5x - 0.2y = 2.4 \\ 0.3x + 0.5y = 0.2 \end{cases}$

d) $\begin{cases} 1.2x + 0.7y = 4.5 \\ -0.4x + 0.3y = -0.7 \end{cases}$

4 Applications of Systems of Equations

Sometimes it is easier to use multiple variables for our unknowns. The general rule is that if we have two unknowns, then we need two equations. Three unknowns, three equations, and so forth.

Example 4.1. Steph Curry made 14 field goals in his last game (2-point and 3-point shots only). From field goals alone, he scored a total of 38 points. How many 2-point shots and 3-point shots did he make?

First Step: State your variables and what they represent.

Second Step: Write your equations to be solved together.

Third Step: Use the appropriate technique (substitution or elimination) and solve for both unknowns.

Fourth Step: State your final answer as a sentence.

Example 4.2. Two integers have a sum of 12. The difference between the larger integer and the smaller integer is 20. What are the two integers?

Example 4.3. If the 3rd term in an arithmetic sequence is 47 and the 8th term is 82, find the common difference and the value of the first term.

Example 4.4. A board 70 cm in length is cut into two pieces. One piece is 8 cm shorter than three times the length of the other piece. Find the length of the two pieces.

Example 4.5. I have only five cent and ten cent coins in my pocket. I have 24 coins altogether, and their total value is \$1.55. How many of each type of coin do I have?

Example 4.6. During the day at the movie theatre 32 people showed up to the afternoon show. If they charged \$5 for students and \$9 for adults and made \$180 in admission fees, how many adults and students were at the show?

Example 4.7. At Lisa's sandwich shop, two chicken sandwiches and four tuna sandwiches cost \$18. Five chicken sandwiches and six tuna sandwiches cost \$34. Find the cost of each sandwich.

Practice Exercises

1. The difference between two numbers is 9. The larger number is 3 more than twice the smaller number. Find the numbers.

2. The sum of two numbers is 3, and twice the larger number is 36 more than three times the smaller number. Find the numbers

3. Gary had a total of $260 in five dollar bills and ten dollar bills. If he has 33 bills in total, how many of each bill does he have?

4. You have 98 cents in your pocket made up of only pennies and nickels. If you have 42 coins in all, how many of each type do you have?

5. Five pencils and four pens cost \$6.15. Three similar pencils and eight similar pens cost \$9.85. How much would you expect to pay for a set of eight pencils and seven pens?

6. In a hockey arena, a seat at rink level costs three times as much as the seat in the upper level. If five seats at the rink level costs \$112 more than eight seats in the upper level, find the cost of the seat at rink level?

7. The perimeter of a rectangle is 40 meters. The width is 4 meters less than the length. Find the dimensions of the rectangle.

8. A rectangle is to be drawn with perimeter 64 cm. If the length is to be 14 cm more than the width, determine the area of the rectangle.

9. Using linear systems, solve for a and d if the following terms are part of an arithmetic sequence.

   a) $a_5 = 21, \quad a_{10} = 41$

   b) $a_4 = -9, \quad a_{15} = -31$

5 Reasoning and Analyzing

Each statement is either **always true, sometimes true,** or **never true**. When you decide on your answer, you must provide a **justification**.

If you choose **sometimes true**, you must show one scenario when it is true, and another, when it is false.

For **always true** and **never true**, you must provide a brief explanation why. Sometimes this can simply be a definition, or sometimes it may require some more mathematical reasoning.

Example 5.1. For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

a) A system of two linear equations has exactly two solutions.

b) If two lines have the same y-intercept, the system has exactly one solution.

c) If two lines have the same slope, then they will never intersect.

d) For distinct non-zero integers A, B, C, D, E, F, the lines $Ax + By = C$ and $Dx + Ey = F$ will intersect at exactly one point.

e) A system of equations consisting of $y = mx + b$ and $y = nx + c$ has no solutions if $m = n$ and $b \neq c$.

Example 5.2. A student was trying to solve the following system by substitution. Identify which step the student first made the mistake, and explain the error.

$$\begin{cases} y = 2x - 16 \\ 3x - 2y = 13 \end{cases}$$

$$\begin{aligned} 3x - 2(2x - 16) &= 13 && \text{Step 1} \\ 3x - 4x - 32 &= 13 && \text{Step 2} \\ -x - 32 &= 13 && \text{Step 3} \\ -x &= 45 && \text{Step 4} \\ x &= -45 && \text{Step 5} \\ y &= 2(-45) - 16 = -106 && \text{Step 6} \\ \therefore &\textit{answer}\ (-45, -106) && \text{Step 7} \end{aligned}$$

Example 5.3. In each of the following, one item doesn't belong...

a) Which system of equations doesn't belong? Explain.

i. $\begin{cases} y = 2x + 1 \\ y = 2x - 3 \end{cases}$

ii. $\begin{cases} y = x \\ y = x + 5 \end{cases}$

iii. $\begin{cases} 2x - y = 4 \\ 4x - 2y = 8 \end{cases}$

iv. $\begin{cases} y = -3x + 2 \\ y = -3x \end{cases}$

b) Which system of equations doesn't belong? Explain.

i. $\begin{cases} x = 5 \\ y = 2 \end{cases}$

ii. $\begin{cases} x + y = 7 \\ x - y = 3 \end{cases}$

iii. $\begin{cases} 2x + 2y = 14 \\ 2x - 2y = 6 \end{cases}$

iv. $\begin{cases} y = 2x - 3 \\ y = -x + 6 \end{cases}$

Practice Exercises

1. For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

   (a) If you multiply one equation in a system by a non-zero constant, the solution to the system changes.

   (b) A system of two linear equations has no solutions if both equations have the exact same x-intercept and the exact same y-intercept (assume the intercepts are not at the origin).

   (c) For non-zero and distinct numbers a, b, c, d, the lines $y = \frac{a}{b}x$ and $y = \frac{c}{d}x$ intersect at the origin.

   (d) For non-zero constants A, B, C, the system of equations $Ax + By = C$ and $2Ax + 2By = 2C$ has exactly one solution.

   (e) If solving a system by elimination results in the algebraic statement $0 = 5$, the system has infinite solutions.

   (f) For non-zero parameters m, A, B, if the lines $y = mx + b$ and $Ax + By = C$ have no solutions, then $m = -\frac{A}{B}$.

   (g) If a system consists of one horizontal line and one vertical line, it has exactly one solution.

2. A student solved the following system using the elimination method. Identify the step where the error first occurred, explain the conceptual mistake, and provide the correct solution.

$$\begin{cases} 5x - 2y = 24 \\ 3x - 2y = 16 \end{cases}$$

$$\begin{aligned} (5x - 2y) - (3x - 2y) &= 24 - 16 && \text{Step 1} \\ 5x - 2y - 3x - 2y &= 8 && \text{Step 2} \\ 2x - 4y &= 8 && \text{Step 3} \\ 2x &= 8 + 4y && \text{Step 4} \\ x &= 4 + 2y && \text{Step 5} \end{aligned}$$

3. In each of the following, one item doesn't belong. Identify which one doesn't belong and explain your mathematical reasoning.

   (a) Which system of equations doesn't belong? Explain based on their graphical properties.

   i. $\begin{cases} y = 3x - 2 \\ y = 3x + 5 \end{cases}$

   ii. $\begin{cases} 2x - y = 4 \\ 4x - 2y = -8 \end{cases}$

   iii. $\begin{cases} y = \frac{1}{2}x + 1 \\ x - 2y = -2 \end{cases}$

   iv. $\begin{cases} 5x + 5y = 10 \\ x + y = 5 \end{cases}$

   (b) Which system of equations doesn't belong? Explain based on the most efficient algebraic strategy to solve them.

   i. $\begin{cases} x = 2y + 5 \\ 3x - 4y = 10 \end{cases}$

   ii. $\begin{cases} 4x - 3y = 12 \\ 5x + 3y = 6 \end{cases}$

   iii. $\begin{cases} y = -3x + 1 \\ 2x + 5y = -8 \end{cases}$

   iv. $\begin{cases} -2x + y = 7 \\ 4x - 3y = 11 \end{cases}$

6 Answer Key

Section 1 - Solving Systems by Graphing

1. a) $(-1, 2)$ b) $(3, 4)$
2. a) $(1, 1)$ b) $(-3, -1)$ c) $(-1, -4)$
   d) $(1, -2)$ e) $(-6, 6)$ f) $(0, 1)$
3. Answers may vary: Approx $(-0.8, 1.6)$. Using technology, $(-8.57, 1.57)$
4. a) $(4, 5)$ b) $(-4.5, -4.5)$ c) $(1.4, 0.4)$
   d) $(0.875, -2.5)$ e) $(8, 1800)$ f) $(1.98, 2.14)$

Section 2 - Solving Systems by Substitution

1. a) $(-1, 1)$ b) $(6, -2)$ c) $(-\frac{1}{3}, \frac{4}{3})$ d) $(-4, -3)$
   e) $(\frac{5}{2}, -\frac{1}{2})$ f) $(\frac{40}{3}, \frac{32}{3})$ g) Infinite Solutions h) No solutions.
2. a) $(3, 2)$ b) $(-2, -6)$ c) $(\frac{28}{29}, \frac{98}{29})$ d) $(1, 1)$
3. a) $(\frac{18}{5}, \frac{84}{5})$ b) $(\frac{88}{13}, \frac{105}{13})$ c) $(6, 3)$ d) $(\frac{122}{63}, \frac{-52}{21})$

Section 3 - Solving Systems by Elimination

1. a) $(2, 6)$ b) $(3, -6)$ c) $(3, 2)$ d) $(\frac{24}{13}, \frac{7}{13})$
   e) $(\frac{3}{2}, 1)$ f) $(2, 0)$ g) $(3, 2)$ h) $(1, -\frac{2}{3})$
2. a) $(3, -1)$ b) $(4, -3)$ c) $(6, 3)$ d) $(-2, 4)$
3. a) $(\frac{204}{25}, \frac{222}{25})$ b) $(\frac{-30}{47}, \frac{168}{47})$ c) $(4, -2)$ d) $(\frac{23}{8}, \frac{3}{2})$

Section 4 - Applications of Systems of Equations

1. The numbers are 15 and 6.
2. The numbers are 9 and -6.
3. There are 14 five dollar bills and 19 ten dollar bills.
4. There are 28 pennies and 14 nickels.
5. Eight pencils and seven pens would cost \$10.50.
6. The cost of a rink level tickets is \$48.
7. The width is 8 m and the length is 12 m.
8. The area is 207 m^2.
9. a) $a = 5, d = 4$ b) $a = -3, d = -2$

Section 5 - Reasoning and Analyzing

a) Always true. When solving for x, we get $x = \dfrac{c+b}{a}$, and since all values are positive, then we must have a positive value for x.

b) Sometimes true. It will depend on the values of b and c. If c is smaller than b, then the answer is a negative number. If c is greater than b, then the answer is positive. In fact, if c and b were equal, then the answer is zero (neither positive nor negative).

c) Never True. Because ax is the same on both sides, the x terms will cancel. This will leave an equation of $b = c$, which can only be true or false. So either infinite solutions, or none.

d) Sometimes true. If $a \neq c$, then there is exactly one solution. However, if $a = c$, then there are no solutions, because b cannot be zero.

e) Always true. $(0, 0)$ is on both lines, regardless of the values of a, b, c, d.

f) Never true. The slope of the line $Ax + By = C$ is $m = -\dfrac{A}{B}$.

g) Always true. The slope of the line $Ax + By = C$ is $m = -\dfrac{A}{B}$.